

CSCE 671 HUMAN-COMPUTER INTERACTION · MODULE 09 · LECTURE NOTES

Qualitative Methods

Finding out what you did not know to ask.

MODULE OUTCOMES

- Explain what qualitative methods establish.
- Conduct an interview that produces usable data.
- Explain contextual inquiry and observation.
- Explain why what people say differs from what they do.
- Analyse qualitative data systematically.

1 What qualitative methods are for

QUALITATIVE METHODS GENERATE HYPOTHESES AND EXPLAIN MECHANISMS; THEY DO NOT MEASURE RATES

An experiment can tell you that design A is faster than design B; it cannot tell you why, or what else the participants were actually trying to accomplish — which is exactly what qualitative work is for, and is frequently the more useful finding.

And it cannot tell you anything about the question you did not think to ask — which is the single largest risk in any study, and the reason to look before you measure.

So the sequence is: observe to find out what matters, then measure the thing that turned out to matter — and **measuring first means measuring whatever you happened to think of**, which may not be the variable that explains anything.

Which makes the two kinds of method complementary rather than competing: 'eight of twelve people did X' from a qualitative study is not a rate, and is an excellent reason to go and measure one (§4's closing callout).

2 Interviews

ASK ABOUT THE PAST AND THE CONCRETE

"Tell me about the last time you did this."
 "Walk me through what you did yesterday."
 "Show me."

DO NOT ASK ABOUT THE FUTURE OR THE HYPOTHETICAL

"Would you use a feature that..."
 "How often do you think you would..."
 -- people are poor at predicting their own
 behaviour, and worse at predicting it to
 somebody who clearly wants a yes.

AND DO NOT ASK FOR SOLUTIONS

"What would you want here?" produces a feature
 request. "What were you trying to do?" produces a
 problem, which is the thing you can design
 against.

THE RULE: ask for behaviour, not for opinion or
 prediction. And then ask "why" until it stops being
 informative.

Ask for behaviour, not for opinion or prediction — which is the single rule that makes an interview worth running, and **the past-and-concrete formulation is the whole technique**. The reason the solution question fails is worth stating: **a feature request encodes the person's model of what is possible**, which is narrower than yours, whereas **a problem statement leaves the solution space open** (Module 04 §4's 'sometimes the user's model is better' is the exception, and it is about the domain rather than about the interface).

AND A LEADING QUESTION GETS THE ANSWER YOU LED TOWARD

'Was that confusing?' gets a yes; 'what were you thinking there?' gets a description — and **only the second one is data**, because the first supplied both the category and the valence.

And the participant wants to be helpful, which means **they will agree with whatever you appear to believe** rather than contradict somebody who has given up an hour to talk to them (Module 08 §4's demand characteristics, in a conversational setting where they are stronger).

So keep your own hypothesis out of the room — do not state it, do not hint at it, and do not react differently to confirming and disconfirming answers — and **be willing to sit through silence, which is where the unprompted material arrives**.

Which is the same discipline as Module 07 §1's — **the interviewer's job, like the facilitator's, is to be uninteresting** — and it is equally difficult in both settings and for the same reason: the natural social instinct is to be responsive.

3 Observation

PEOPLE CANNOT ACCURATELY REPORT THEIR OWN BEHAVIOUR, WHICH IS NOT DISHONESTY

Self-reported frequencies are **unreliable** in a measured and systematic way, **workarounds are forgotten** because they have become automatic, and **the steps of a familiar task cannot be enumerated by the person who performs them** — expertise is exactly the compiling-away of the steps (Module 03 §4).

So the workaround is the finding, and they will not mention it — the spreadsheet kept open beside the application, the note stuck to the monitor, the value re-typed because it cannot be copied, the report exported and fixed by hand every Monday.

Which is precisely why contextual inquiry exists: watch them do the real work in the real place, and ask about what you can see rather than about what they can recall.

And the physical environment is itself data — what is written down, what is printed out, what is kept in a second window, what is on a sticky note — all of which record something the software failed to provide and which nobody filed a bug about.

- **Go where the work actually happens**, because **the environment is half the data** — the interruptions, the other people, the phone, the second screen, and the noise are all part of the conditions the design has to work in.
- **Adopt the apprentice stance: 'teach me how to do this'** — **which makes their expertise the subject and licenses you to ask obvious questions** without either party being embarrassed, and is a far better frame than 'evaluate this'.
- **Record the artefacts** — the spreadsheets, the handwritten notes, the printouts, the bookmarks — since **each one is a feature request nobody filed** and is concrete evidence rather than an opinion.
- **Note the interruptions and how they recover from them**, which **no laboratory study will ever show you** (Module 03 §1) and which may dominate the real experience of using the software.
- **And ask about the exceptions** — the unusual customer, the month-end case, the thing that went wrong last week — **because the unusual case is where the design usually breaks** and where the workarounds concentrate. **Every artefact beside the software is a feature request nobody filed**, which makes **a desk the cheapest requirements document available**.

4 Analysis

- **Transcribe, or at the very least index, the sessions**, because **working from memory selects for whatever was most memorable** — the vivid complaint, the articulate participant — **rather than for whatever was most common**.
- **Code the data: label every segment with a category**, and **let the categories emerge from the material rather than imposing a scheme you brought** (CSCE 638 Module 10 §4's same requirement for error analysis).

- **Then count the codes**, so that you know what recurred and how often within your sample — **which is not a population rate and is considerably better than an impression**, and is reportable as what it is.
- **Have a second person code a sample independently and compare**, because **a coding scheme that only you can apply is not a finding** — it is an interpretation, and the agreement rate is the evidence that it is more than that.
- **And look specifically for the disconfirming case** — the participant who did not do the thing, the session that contradicts the pattern. **Looking for the disconfirming case is what distinguishes analysis from illustration: a report of five supporting quotations is a selection, not a finding**, and the reader cannot tell how many contradicting ones were available.

AND WHAT A QUALITATIVE FINDING HONESTLY SUPPORTS

'This mechanism exists and here is how it works' — **which is a strong and genuinely useful claim**, and is the one these methods are designed to support. Establishing that something happens at all, and by what route, is valuable and does not require a sample.

'Eight of our twelve participants did X' — **which is a description of twelve people and is not an estimate of a population**, however much it reads like one.

And not 'X% of users do this', which **requires a sample designed for the purpose** (Module 08) and a sampling frame you almost certainly do not have.

So report the number of participants, how they were recruited, and the mechanism rather than the rate — **which is Module 13's claim form and is what makes a qualitative study citable** rather than merely suggestive.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Beyer & Holtzblatt — Contextual Design	§3's method in full — the apprentice stance and the artefact collection, from its originators. Library copy.
Portugal — Interviewing Users	§2 practically — short, and the question-phrasing advice is the best available. Library copy.
Lazar, Feng & Hochheiser, the qualitative chapters	§4's analysis — coding, agreement, and what the results support. Library copy.
Nisbett & Wilson — Telling more than we can know (free)	§3's finding in the original — the classic result on the unreliability of introspective reports.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write ten interview questions** and classify each as behavioural, hypothetical, or solution-seeking.
- 2 **Rewrite the bad ones.**
- 3 **Run one interview** and count your own leading questions afterwards.
- 4 **Sit through ten seconds of silence** deliberately and note what arrives.
- 5 **Watch somebody do real work** in their own environment for an hour.
- 6 **Photograph every artefact** on their desk and classify each as a feature request.
- 7 **Ask them to describe the task first**, then watch, and compare the two accounts.
- 8 **Code one session** with categories drawn from the material.
- 9 **Have somebody else code a sample** and compute your agreement.
- 10 **Find a disconfirming case** for your main finding, and report it.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What do qualitative methods establish, and what do they not?
- 2 Why observe before measuring?
- 3 Give the rule about interview questions, with examples of each kind.
- 4 Why does asking for solutions fail?
- 5 Why does a leading question waste a session?
- 6 Why can people not report their own behaviour?
- 7 What is the finding they will not mention?
- 8 Give five rules for observation, and what the artefacts are.
- 9 Give five rules for analysis, and which separates it from illustration?
- 10 What does a qualitative finding support, and what does it not?